

Five small models, one machine, two jobs

A plain account of the round.

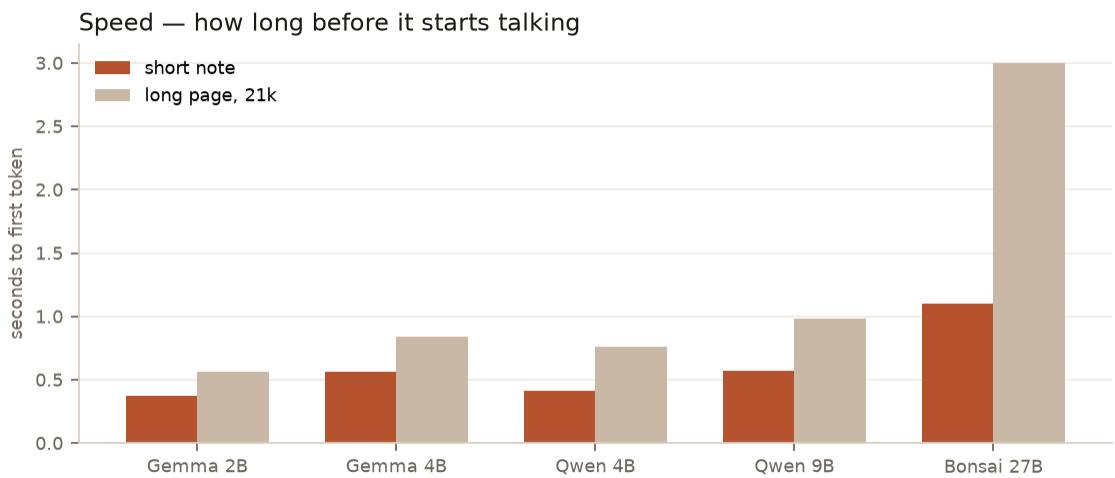
23 September 2026

Five small models, one machine, two jobs. The numbers below are the round; the pictures sit beside the sentences they belong to.

Speed

How fast each one starts talking, and how fast it writes, on short notes and on long ones.

Model	Short first token	Decode, short	First token at 21k
Gemma 2B	0.37 s	38 tok/s	0.56 s
Gemma 4B	0.56 s	23 tok/s	0.84 s
Qwen 4B	0.41 s	39 tok/s	0.76 s
Qwen 9B	0.57 s	22 tok/s	0.98 s
Bonsai 27B	1.10 s	20 tok/s	3.0 s



Seconds to the first token. Dark bar: a short note. Pale bar: a long page.

Qwen 4B and Gemma 2B are the quick ones. Bonsai is fine on a short note and painful on a long page.

Gemma 4B takes longer to begin its answer than its first-token measure suggests: the first answer token arrives at a 3.4 s p50 on a short note and 4.1 s p50 at about 22k tokens (n=5).

Pull fields out of a notice

One hundred and fifty fake letters. The score is how many of the dates, amounts, and actions it got right.

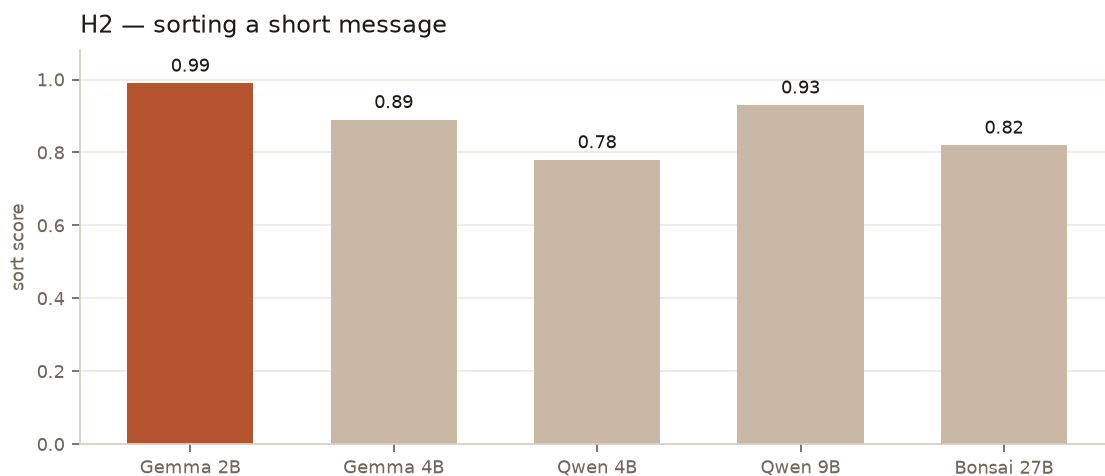
Model	Field score	Dates exact	Made-up fields
Gemma 2B	0.64	0.67	0.26
Gemma 4B	0.58	1.00	0.32
Qwen 4B	0.31	0.15	0.63
Qwen 9B	0.64	1.00	0.23
Bonsai 27B	0.44	0.61	0.46

Gemma 2B and Qwen 9B tie on the main score. Qwen 9B is the one that gets every date. Qwen 4B invents more than it keeps.

Sort a short message

Three hundred messages, twelve bins, including "none of these" and "unclear."

Model	Sort score	Knew when to abstain	Sure and wrong
Gemma 2B	0.99	0.96	0.01
Gemma 4B	0.89	0.50	0.00
Qwen 4B	0.78	0.16	0.13
Qwen 9B	0.93	0.62	0.03
Bonsai 27B	0.82	0.58	0.10



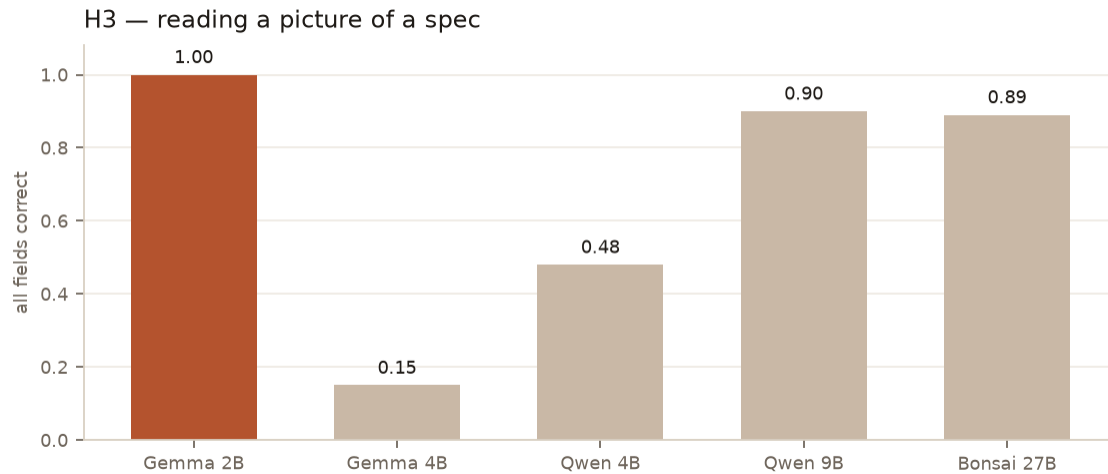
Sort score across twelve bins.

Gemma 2B almost never mis-files, and it is the only one that reliably says it does not know. Qwen 4B guesses.

Read a picture of a spec

One hundred and fifty rendered pictures. Did it get every field, and did it invent a colour that was not there?

Model	All fields	Read the text	False presence
Gemma 2B	1.00	1.00	0.00
Gemma 4B	0.15	1.00	0.85
Qwen 4B	0.48	0.99	0.27
Qwen 9B	0.90	0.90	0.00
Bonsai 27B	0.89	0.89	0.00



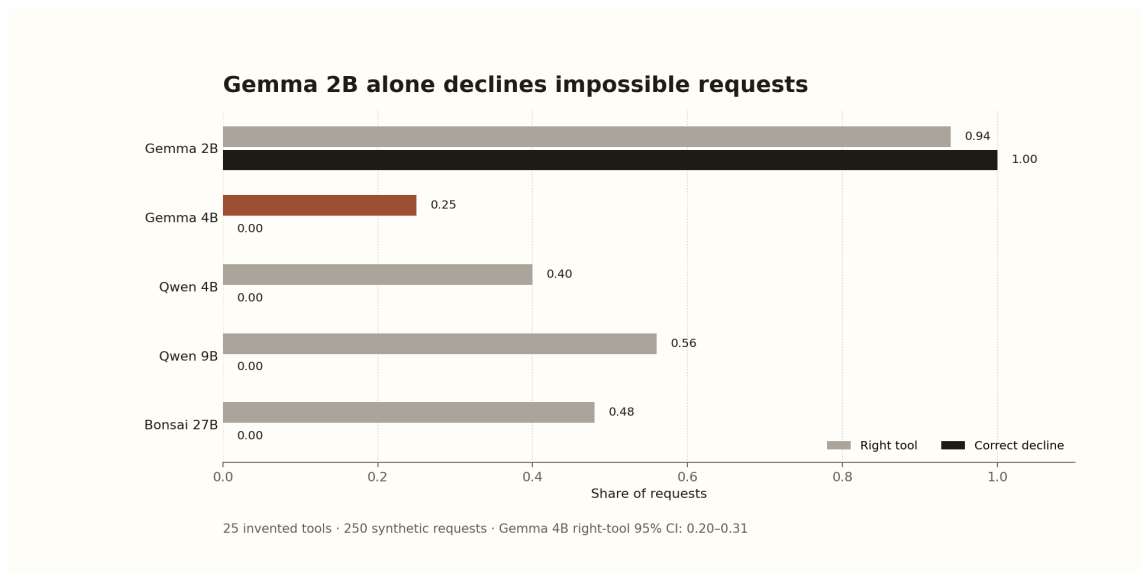
Share of pictures where every field was right.

Gemma 2B read every picture cleanly. Gemma 4B can read the words and then invents shapes that are not in the picture, most of the time. Qwen 9B and Bonsai are close behind and do not invent.

Call the right tool

Two hundred and fifty requests, a fake catalogue of twenty-five tools. The number that matters for a backup is whether it refuses a job that cannot be done.

Model	Right tool	Decline when impossible
Gemma 2B	0.94	1.00
Gemma 4B	0.25	0.00
Qwen 4B	0.40	0.00
Qwen 9B	0.56	0.00
Bonsai 27B	0.48	0.00



Correct refusals. Four models never declined.

Only Gemma 2B both picks the tool and refuses the impossible ones. The other four never declined. When they did call a tool, their arguments were exact, including Gemma 4B.

Keep six rules

Sixty conversations, four turns each. Rules: one short sentence, JSON, do not say a secret word, Spanish, one calm marker, a fixed sign-off.

Model	Rules kept	Held under attack
Gemma 2B	0.16	0.14
Gemma 4B	0.17	0.17
Qwen 4B	0.92	0.81
Qwen 9B	0.93	0.83
Bonsai 27B	0.77	0.59

The two Qwens follow the format. Gemma 4B never says the secret word, but its mathematical fence around the JSON costs it the format score. Gemma 2B breaks the style on nearly every turn and still only leaks the secret eight times. The score punishes the style, not the leak.

Answer from a long page

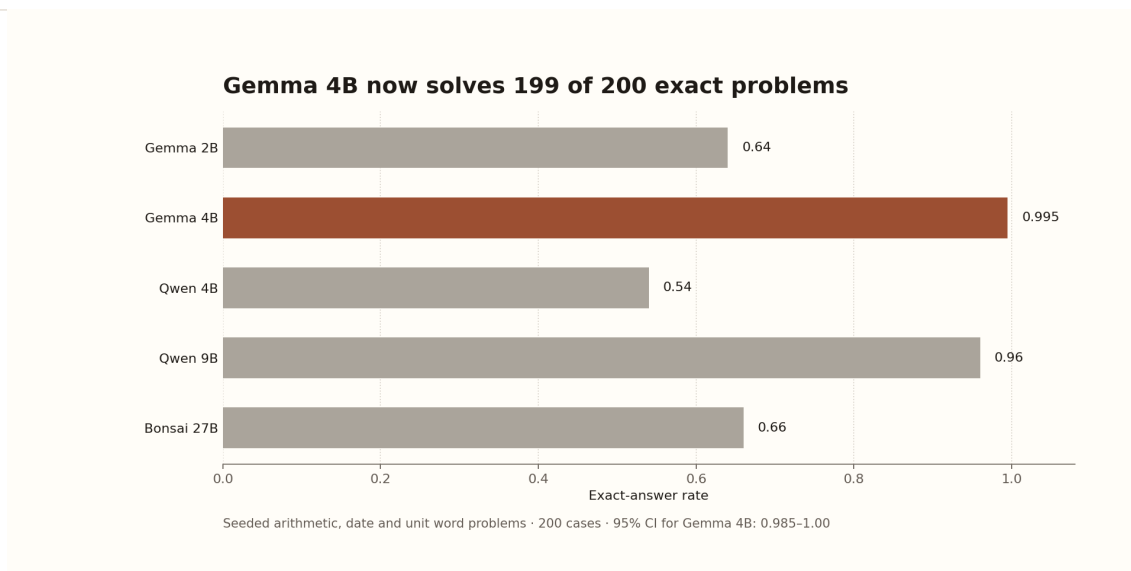
One hundred questions, the fact buried at four depths. Every model scored 1.00 at every depth, and none invented an answer when the page did not have one. Bonsai was the least sure of a correct answer (error 0.20); the others were sure and right.

The long-page test did not separate anyone. The fact sat near the start of the page at every labelled depth, so a perfect score here does not mean they can find a fact at the end of a real document.

Sums, dates, units

Two hundred exact-answer problems.

Model	Exact
Gemma 2B	0.64
Gemma 4B	0.995
Qwen 4B	0.54
Qwen 9B	0.96
Bonsai 27B	0.66



Exact answers. Gemma 4B leads the five models.

Gemma 4B is the calculator: it solved 199 of 200, just ahead of Qwen 9B at 0.96.

Same answer twice

Thirty prompts, repeated at temperature zero. Every model repeated itself exactly (1.00). Asked to emit JSON while warm, only Qwen 9B was reliable (0.97). Gemma 2B produced none; Gemma 4B produced valid JSON 0.07 of the time (95% CI 0.00–0.17).

What I would pick

Helper: Gemma 2B. It sorts, it reads pictures, it knows when to abstain, and it is fast. Call Gemma 4B when the job is a sum and there is time to wait: it never declines, invents shapes in pictures, and takes three to four seconds to begin its answer. Qwen 9B is the better reader of dates and the one to call when the work demands a strict format.

Backup that must decline: Gemma 2B, because it is the only one that declines. Backup that must follow a format: Qwen 9B, and accept that it will attempt a tool call it should have refused.

What surprised me: the bigger Gemma is worse than the smaller one at several helper tasks, yet it is a near-perfect arithmetic solver. Bonsai's size bought neither speed nor a refusal. And the long-page test, which was meant to be the hard one, told us nothing — five perfect scores on a page that hid the answer in the same place every time.

Sources

1. Gemma 4B evidence: TABLE.md

2. Round evidence and charts: present/round-02/
3. Synthetic seeded fixtures and mechanical scorers.

— *Davidson*